INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

ASSIGNMENT

Assignment Title: Linux System Administraton Adventure

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Course: Foundation of Linux, Networking and Bash Scripting
Course Code: BVWS101

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26/OCT/2025

Part 1: Research & Analysis - The Attack of the Layers

1. Layer: **Layer 7 – Application**
    - Attack Name: Phishing
    - How the Attack Works: Phishing is a social engineering attack where cybercriminals impersonate trusted entities like banks or email providers to trick users into revealing sensitive information such as passwords or credit card numbers. These attacks often come through fake websites or deceptive emails.
    - Security Control/Mitigation: To reduce the risk, organizations should use email filtering tools, train employees to spot suspicious messages, and enforce multi-factor authentication.
2. Layer: **Layer 2 – Data Link**
    - Attack Name: ARP Spoofing
    - How the Attack Works: ARP Spoofing manipulates the Address Resolution Protocol by sending false ARP messages on a local network. This tricks devices into associating the attacker's MAC address with a legitimate IP address, allowing them to intercept or redirect traffic.
    - Security Control/Mitigation: Network switches can be configured with Dynamic ARP Inspection (DAI), and static ARP entries can be used to prevent spoofing.
3. Layer: **Layer 4 – Transport**
    - Attack Name: TCP SYN Flood
    - How the Attack Works: This attack floods a server with TCP SYN requests but never completes the handshake. The server keeps resources allocated for these half-open connections, eventually becoming overwhelmed and unable to respond to legitimate traffic.
    - Security Control/Mitigation: Using SYN cookies and configuring firewalls to limit incomplete connections can help defend against this type of attack.

Part 2: OSI Layer Security Reference Tables

OSI Layer Number	**Layer Name**	**Key Function**	**Primary Cybersecurity Focus (What are we protecting at this layer?)**
7	Application	Provides network services directly to user applications (e.g., HTTP, FTP, DNS).	Protecting the applications and data from malicious input and unauthorized access.

6	Presentation	Translates data formats and handles encryption and compression.	Ensuring secure data encoding and protecting against format-based exploits.
5	Session	Manages and maintains connections between applications	Securing session integrity and preventing hijacking or unauthorized access.
4	Transport	Ensures reliable data transfer with error checking and flow control.	Preventing DoS attacks and securing end-to-end communication.
3	Network	Handles routing and forwarding of data across networks using IP addresses.	Protecting against spoofing, routing attacks, and ensuring secure data delivery.
2	Data Link	Manages MAC addressing and error detection for data frames.	Preventing MAC spoofing and ensuring frame-level integrity.
1	Physical	Transmits raw bits over physical media like cables and wireless signals	Securing physical infrastructure and preventing unauthorized physical access.

Table 2: Protocol & Technology Mapping

OSI Layer Number	**Layer Name**	**Example Protocols & Technologies**	**Cybersecurity Technology/Tools**

7	Application	HTTP, HTTPS, DNS, FTP, SMTP	Web Application Firewall (WAF), Data Loss Prevention (DLP) systems
6	Presentation	SSL/TLS, JPEG, MPEG, ASCII	TLS encryption, secure encoding tools
5	Session	NetBIOS, RPC, PPTP	Session management tools, token-based authentication
4	Transport	TCP, UDP	Firewalls, Intrusion Detection Systems (IDS)
3	Network	IP, ICMP, IPSec	Routers with ACLs, VPNs, anti-spoofing filters
2	Data Link	Ethernet, PPP, ARP	MAC filtering, Dynamic ARP Inspection (DAI)
1	Physical	DSL, USB, Bluetooth, Ethernet cables	Physical locks, surveillance, port security

Table 3: Attack & Mitigation Mapping

OSI Layer Number	**Layer Name**	**One Specific Cyber Attack**	**One Appropriate Mitigation Strategy**
7	Application	SQL Injection	Input Validation & Prepared Statement
5	Session	Session Hijacking	Use encrypted sessions and secure token handling.

6	Presentation	SSL Stripping	Enforce HTTPS and use HSTS headers.
1	Physical	Cable Tapping	Physical security controls and port access locks